Mission Control Interview – Operational Protocols & Communication (Flight Controller)

Overview

This document captures firsthand insights into mission control operations, communication protocols, and decision-making practices for flight controllers.This data has been compiled from interview data from four real flight controllers. Through our interviews we have been able to gain insights from the following Flight Controller positions of FIDO, ETHOS, EVA and CRONUS.

FIDO

Role Description-

FIDOs are responsible for orbital mechanics and trajectory design for astronaut spaceflight missions, including launch targeting, engagement with the ISS, lunar flybys, and deorbit planning. FIDOs also monitor trajectory health, calculate burn plans, and advise the Flight Director on maneuver and reentry. It also involves tool and software development, crew training, and regular high-fidelity mission simulations to rehearse normal and abnormal scenarios.

Decision Making-

FIDO decision making is based on orbital mechanics models, real time tracking data, fuel levels, and vehicle constraints. Since it's very technical, they need to quickly plan out whether or not something is possible- all while prioritizing crew safety. Regardless of this, the Flight Director ultimately has the final say.

Communication -

FIDOs constantly communicate with the Flight Director and officers/specialists related to a situation. They usually convey information in emotionless, quantifiable manners.

ETHOS

Role Description-

ETHOS are flight controllers responsible for monitoring and managing a spacecraft's life-support and environmental control systems. They oversee electrical power levels, thermal regulation, cooling loops, water recovery, cabin air circulation, and fire detection/suppression systems.

Decision Making-

ETHOS decision making is very analytical and procedure based, requiring quick and calm thinking under pressure. They must consider the crew's safety, vehicle integrity, and impact on the mission. These choices are the result of coordinating with other flight controllers, making critical calls to the flight director, and support recommendations with rationale.

Communication -

ETHOS generally does not speak directly to the astronauts. They report status updates and anomalies to the flight director, who makes the final call on mission-level decisions.

EVA

Role Description-

EVAs are the flight controller responsible for planning, supporting, and monitoring spacewalks for astronauts, managing emergency scenarios and tracking vitals.

Decision Making-

When making decisions, EVAs are focused on safety first and driven by margins and procedures- however, they are also very adaptable. They must think ahead and prepare for major risks.

Communication -

EVA communication style is tightly coordinated, step specific, and in constant contact with Mission Control and routes guidance through CAPCOM.

CRONUS

Role Description-

CRONUS is responsible for the spacecraft's communication systems and onboard data networks. They manage radio frequency links between the vehicle and ground satellites, command video systems, and ensure the flow of telemetry data to all other flight control consoles

Decision Making-

Decision making is centered on link availability and data integrity. They must prioritize bandwidth usage and anticipate "acquisition of signal" (AOS) times. They make rapid adjustments to antenna pointing and transmitter settings to prevent or recover from a loss of signal.

Communication -

CRONUS serves as the technical gatekeeper for the "pipes" of information. They proactively alert the Flight Director and CAPCOM of upcoming communication "blind spots" (Loss of Signal) to ensure critical voice instructions to the crew are not interrupted.

ISO

Role Description-

The Inventory Stowage Officer (ISO) is responsible for the real-time tracking of United States (U.S.) inventory onboard the ISS. The ISO manages a five-part database called the Inventory Management System (IMS), which keeps track of all the items on the ISS and their stowage locations. The IMS database servers are located at five prime databases: onboard the ISS, and in MCC-Houston, MCC-Moscow, Tsukuba in Japan (Japanese Control Center) and COL-CC (European Control Center). Just before a Progress (a Russian uncrewed supply craft) launches, a sixth server is linked into the IMS at Baikonur, Kazakhstan.

PLUTO

Role Description-

The Plug-in-Plan and Utilization Officer (PLUTO) maintains and coordinates changes to the U.S. segment of the electrical Plug-in Plan (PiP), which tracks the organization of all on-orbit portable equipment and associated constraints. The PLUTO is also responsible for certain Station Developmental Test Objectives, or SDTOs (testing of equipment) during the mission. This includes testing of the Wireless Instrumentation System (WIS) and also of ROBONAUT activities. The PLUTO Flight Control Team is responsible for the Space Station Computer (SSC) and the Joint Station Local Area Network (JSL).

SPARTAN

Electrical power is one of the most important resources on the International Space Station (ISS). It is used to power lights, fans, motors and scientific research, among many other needs. The Station Power, Articulation and Thermal Control (SPARTAN) flight controller supports the electrical power system on the International Space Station (ISS) by managing the operation of the hardware and software on the United States On-orbit Segment (USOS), the United States' portion of the ISS.

ADCO

Role Description-

The ADCO flight controller plans upcoming orientations and attitude maneuvers, and visiting vehicle dockings to the ISS. This person monitors the ISS position, velocity (speed and direction) and attitude to make sure that they do not change unless commanded to do so by the computers aboard the ISS. The ADCO (pronounced add-co) flight controller works together with a Russian flight controller counterpart to calculate and manage the attitude (or orientation) of the International Space Station (ISS).

OSO

Role Description-

The Operations Support Officer (OSO) plans and coordinates all internal vehicle maintenance activities onboard the ISS. The OSO manages the development, training and analysis of maintenance tasks on the ground prior to carrying the tasks out in orbit and keeps the Flight Director aware of the status of repairs. The OSO is also responsible for the internal mechanisms which attach pressurized modules to the ISS, as well as external mechanisms which attach research cargo and spare pallets (portable platforms on which goods are placed for storage or moving) to the truss.

ISE

Role Description-

Integration and Systems Engineer (ISE) Responsible for monitoring key systems and operations of the visiting vehicles approaching and mating to US Segment of the ISS.

OPS PLAN

Role Description-

The Ops Planner (OPS PLAN) is responsible for developing, coordinating and monitoring execution of the daily schedule of activities on the ISS. The daily activities are organized in a Short-Term Plan (STP), which provides necessary information to carry out each day's activities - both for the crews on the ISS and on the ground. If the day's activities are not completed and need to be rescheduled, the OPS PLAN also provides input to the Long Range Planning (LRP) team.

ROBO

Role Description-

Responsible for the ISS Robotic Arm flight control and analysis. Develops procedures to complete robotic operations and builds computer simulations to design procedures. Supports real time on board training and operations for the ISS astronauts and performs many of the operations via ground commanding from the MCC.

TOPO

Role Description-

Plans ISS trajectory and coordinate with ISS partners as needed. Assesses accuracy of ISS state vector (onboard and ground) and updates as required. Maintains ISS onboard tracking and data relay satellite state vectors. Evaluates trajectory of ISS and associated visiting vehicles for conjunctions and potential debris avoidance maneuvers.

VVO

Role Description-

Provides rendezvous expertise and integration with MCC Houston to assure safe operations as vehicles are approaching or departing from the ISS. Visiting vehicles include SpaceX, Orbital, Japanese HTV, and Russian Soyuz and Progress, as well as the future Boeing CST-100 and other new commercial crew vehicles. Interfaces extensively with the control centers of visiting vehicles.

RIO

Role Description-

Formerly known as the Russian interface officer. Responsible for integrating operations between MCC-Houston and the other International Partner (IP) Control Centers. RIO works closely in conjunction with the Houston Support Group teams located at the IP Control Centers.

Pointing

Role Description-

Performs line-of-sight analysis used in determining science viewing opportunities and communications coverage for the ISS and visiting vehicles. Ensures effective communications coverage as well as other functions dependent on vehicle, antenna, or sensor pointing.

Decision Making

Real-Time Judgment & Mental Readiness

- When you are a flight controller, there is no way to plan for anything.
- Flight Controllers must have basic knowledge for everything they do. They will have to make decisions on the fly based on their training.
- Flight Controllers must be in the moment at all times. There is a high concentration needed.

Communication & Shared Understanding

- There are certain levels of people that Flight Controllers talk to during missions.
- Flight Controllers must understand what their peers need to know when making a decision.

Broader Perspective

- There is a need to always look at the bigger picture.
- There is the fundamental policy to do the right thing when working.
- Operations are the last line of defense. They keep the crew safe.

Process & Complexity

- There is a very rigorous procedure in any decisions that they make as a flight controller.
- Flight controllers must weigh their options to see what they must do. This is in context with the complexity of the ISS, the entire program, the people, and the organizations.

Conflict Resolution

Structure & Chain of Command

- Control rooms have a clear chain of command. With multiple groups supporting them.
- You must be aware of the overall context of the situation and how your role fits as a Flight Controller.
- If the higher command does not choose your decision, get back in line and keep going as a team.

Communication & Recommendation Framing

- Flight Controllers must communicate a preferred path, but describe the alternative.
- Flight Controllers must be clear with the impact, why you are recommending it, and what your time to effect is.
- Flight Controllers talk to backrooms about what-ifs.

Readiness & Response

- As a flight controller, there is a wider spectrum, except for something to happen.
- A flight controller must know immediately how to deal with the problem.
- As a flight controller, you are aware that when situations arise, this is what everyone is trained for, so be ready.
- They must be there right away when things go bad.

Procedures & Technical Management

- Flight Controllers must read procedures.
- The most common technical issues involve day to day management.
- Flight controllers are there 24/7.
- Flight controllers always have something to do.

- Technical hardware over time will break down so you only want to run it for a certain amount of time.
- Flight Controllers must “Micro manage” systems.
- Flight controllers need to be able to fix situations that could have gone worse if they didn't have expertise and knowledge.

Training

Technical Skills vs. Soft Skills

- Technical skills when training as a flight controller are more straightforward than soft skills, which are more human skills.
- Flight controllers must be an expert in their system and anticipate what the astronauts will do next.
- Flight Controllers are aware of FIW (Failure Impact Workaround).
- When training as a flight controller, the better you get at one skillset, the better you are at other systems.

Training Flight Controller Communication

- Flight controllers need to understand how to listen to multiple channels at once and know what to focus on.
- Flight controllers must learn how to communicate in this specific environment.
- Flight controllers must know how to package their calls.

Troubleshooting & Critical Thinking

- Flight controllers must come up with a good trouble shooting process with any situation that arises.
- Flight Controllers must make sure they understand the whole situation.
- Flight Controllers must understand why the failure means it is critical.

Focus, Cognitive Load & Situational Awareness

- Flight Controllers can only focus on one thing at a time, they must be aware of that.
- A flight controller must manage their own CPU, they must not get overwhelmed and not lose focus.
- Flight controllers must have high situational awareness.
- The hardest part when training is to teach flight controllers to stay ahead of the rest of their team.

Training Environment & Skill Development

- During training as a flight controller, a lot comes down to practice and simulations where you are overwhelmed and at your limit.
- Training takes lots of practice as a flight controller.
- Repetition and reinforcement is key when training as a flight controller.

Communication

Information Structure & Procedures

- Flight Controllers might point to a whole paragraph which can be very overwhelming.
- Flight Controllers Add identifiers to each section of data or step in the procedure.

Communication Layers & Message Delivery

- Flight Controllers communicate with Capcom which makes an added layer of the telephone game.
- Flight Controllers pace the timing and package of your message.
- Flight Controllers must hit the main important points.
- Learning how to talk like a flight controller and getting concise info across FAST during operations.
- Communication is one of the biggest issues as a Flight Controller.

- How to communicate a lot of complex information concisely at the correct level for whichever audience (flight director, or other flight controllers).

- Running body language, hearing their Q&A and adjusting your information that you need to share.

- During missions, there is a lot of adapting to situations when communicating with others as a Flight Controller.

- In a situation where “This thing just failed.”Flight controllers need to focus on communication and take in clues, and give them what they need based on contextual clues.

Audience Awareness & Expectations

- A big part is knowing your audience as a Flight Controller.

- Everyone has their own communication preferences.

- Learn how to quickly adapt to people's communication styles.

- Knowing if you have worked with them before and whether there is an understanding of experience level.

- Flight Directors will want the big picture, the flight controller’s recommendation, and for you to have other alternative options available to them.

- Flight Director’s give you a time frame for their expectations and will come to the flight controllers afterwards for their explanations for their responses in high stress situations.

Training, Preparation & Skill Development

- PRACTICE A LOT as a Flight Controller.
- There is engineering support as a Flight Controller.
- The KEY thing is to communicate a lot and ask lots of questions as a Flight Controller.
- Flight Controllers must understand “The Five Hows”, they must ask up to 5 times to see how something will work.
- E.g How to deploy wings, How to apply power and control, How to activate, Etc..
- Soft skills take a lot of practice as a Flight Controller.
- Gets to maintain these soft skills while in ISS.
- Hardest parts of the job and takes practice as a Flight Controller.

Decision Framing & Operational Response

- Dynamic Environment → A lot of things are very fast paced during critical moments as a Flight Controller.
- There is only a little bit of down time when you are a Flight Controller.
- Give a bit more information when you can as a Flight Controller.
- There are different options; everyone has their own recommendations during missions at mission control as a Flight Controller.
- Supporting when something breaks is a part of the flight controller’s responsibility.
- Learn the FIW (Failure Impact Work Around) as a Flight Controller.
- If you don't, say we need to stop the crew to discuss before proceeding as a Flight Controller.

Knowledge & Situational Awareness

- The best thing you can do is “know your stuff” as a Flight Controller.
- If you don't know your system, you can't keep track of all of that at once as a Flight Controller.
- Flight Controllers must not lack awareness to make an effective call
- Knowing your information is best to get quicker at communication as a Flight Controller.

Visibility In Mission Control Room

Viewable Items In Mission Control

This a list of items that are visible to flight controllers when that they utilize during operations:

- Clock
- Communication status w/ vehicle
- Specific vehicle modes
- Ground track map
- Spatial awareness
- The sun, day night cycle
- Thermal condition
- Power conditions
- What is happening with the vehicle
- Timeline for day
- When are the breaks for flight controllers
- Timers going and whether they are ahead or behind of schedule
- When losing comms, want to keep track of the spacecraft

- Live videos from ISS or Starliner (front of the room)

- Caution and warning board

- Different levels of alarms
- Cannot block the view of someone behind you

Overall the needs of a flight controller will depend on the context of the mission and the individual role of the flight controller.

Console-Level Data & Monitoring

- 13000 parameters that ETHOS owns
- Overview display (high level)
- Plotted data (looking at trends)
- Sensors
- Tools that allow flight controllers to look at system changes
- Tools that act as notifications and beep in a Flight controllers ear
- Command histories and make sure they are getting onto the spacecraft
- ETHOS needs to have things configured for the crew so that the crew can do their part
- Documentation procedures

EVA & Communication Visibility

- The role is a bit different from most as a Flight Controller.
- The best sensor for Flight Controllers in EVA is the astronaut themself.
- New AI transcription of the comms from the astronaut during missions.

- Very helpful for replaying what he said or what astronauts said during missions.
- Got to know where we are stepping through and what is next during missions.

Display Configuration & Layout

- Display Configuration & Layout depends on display real estate that you have during missions.
- Most controllers spend most time looking at their own consoles and just glance up at the big board.
- Used to have a standardized way of doing things when the logins would stay between different people during missions.
- Not everyone gets their own log-ins with their own presets during missions.
- Lots of personal preference for displays during missions.
- No one sets their display settings the same during missions.
- People get frustrated with limits on configuration during missions.
- There are different levels of info that people want available to them during missions.
- Flight controllers must adapt the console to show relevant info.
- All control rooms have different monitor layouts during missions.

Distraction & Attention Management

- Busy → minimize teams, emails during missions.
- You need to communicate with the console team, when emailing do not assume they will see their emails as a Flight Controller.
- Don't want anything distracting as a Flight Controller.
- Puts phone in backpack and zips up during missions.